

Analysing Cell Delay and Power Characteristics from a Liberty File

INTRODUCTION:

This experiment demonstrates how to extract and analyse cell delay and power characteristics from a Liberty (.lib) file, which is the standard format used to describe the behaviour of standard cells in digital VLSI. The .lib file provides detailed information such as pin definitions (inputs, outputs, and their directions), timing data (cell rise/fall delays and transition times), and power parameters (including leakage and internal power). In this exercise, we will explore the osu018_stdcells.lib file and focus on analysing delay and power characteristics of commonly used cells such as INV, NAND2, NOR2, and DFF, to understand how these parameters influence digital circuit performance.

OBJECTIVE:

After completing this lab, you will be able to:

- Understand how **cell delay** is modelled in Liberty files.
- Identify **cell_rise**, **cell_fall**, and **transition times**.
- Interpret **power characteristics** such as leakage and internal power.
- Build an observation table summarizing delay and power values.

PROCEDURE:

Step 1: Open The Library

```
gvim osu018_stdcells.lib
```

```
-VirtualBox:~/project$ gvim osu018_stdcells.lib
```

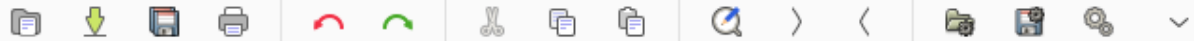
Step 2: Search for a Cell

Use `/cell` (inside `gvim` to find cells.

Example:

```
/cell (INVX1)
```

File Edit Tools Syntax Buffers Window Help



```
"0.047529, 0.043539, 0.052945, 0.059775, 0.082777", \
"0.048555, 0.044674, 0.052352, 0.059345, 0.083031", \
"0.049041, 0.045208, 0.052753, 0.059207, 0.083092");
```

```
}
}
```

```
/* ----- */
* Design : INVX1 *
/* ----- */
cell (INVX1) {
  cell_footprint : inv;
  area : 16;
  cell_leakage_power : 0.0221741;
  pin(A) {
    direction : input;
    capacitance : 0.00932456;
    rise_capacitance : 0.00932196;
    fall_capacitance : 0.00932456;
  }
  pin(Y) {
    direction : output;
  }
}
cell (INVX1)
```

Step 3: Locate Delay Information

Inside the `timing()` block, check for:

```
timing() {
  related_pin : "A";
  cell_rise (table_2d) { values(...); }
  cell_fall (table_2d) { values(...); }
  rise_transition (table_2d) { values(...); }
  fall_transition (table_2d) { values(...); }
}
```

This gives the **delay and transition times** of output pin relative to input pin.

Step 4: Locate Power Information

Check for **internal power** and **leakage power**:

```
leakage_power : 2.5e-06;
internal_power() {
  rise_power (table_2d) { values(...); }
```

```
fall_power (table_2d) { values(...); }
}
```

- leakage_power = static power consumption.
- rise_power/fall_power = dynamic internal power.

Step 5: Repeat for Multiple Cells

Perform Steps 3 and 4 for **INVX1**, **NAND2X1**, **NOR2X1**, and **DFFPOSX1** to compare both combinational and sequential cells.

OBSERVATION:

Cell Name	Delay Parameters	Power Parameters
INVX1	cell_rise, cell_fall, rise/fall transition tables	leakage_power, internal_power (rise/fall)
NAND2X1	cell_rise, cell_fall for A→Y and B→Y	leakage_power, internal_power
NOR2X1	cell_rise, cell_fall for A→Y and B→Y	leakage_power, internal_power
DFFPOSX1	clk→Q delay, setup/hold constraints	leakage_power, internal_power

OUTPUT:

- Extracted **cell delay** data (rise/fall + transition).
- Extracted **leakage and internal power**.
- Summarized results in an observation table.

CONCLUSION:

This experiment successfully demonstrated how to analyse cell delay and power characteristics from a Liberty file. By studying osu018_stdcells.lib, we observed how timing arcs describe gate delays and how leakage/internal power are modelled. Such information is crucial for synthesis, timing analysis, and power estimation, ensuring realistic digital design implementation.